

$$F(y) = \frac{1}{\pi} \arctan\left(\frac{y-y_0}{\gamma}\right) + \frac{1}{2}$$

$$\pi \left[F(y) - \frac{1}{2} \right] = \arctan\left(\frac{y-y_0}{\gamma}\right)$$

$$\tan\left[\pi \left(F(y) - \frac{1}{2} \right)\right] = \frac{y-y_0}{\gamma}$$

$$y_0 + \gamma \tan\left[\pi \left(F(y) - \frac{1}{2} \right)\right] = y$$

Per tant:

$$\pi(y - \frac{1}{2})$$

$$F^{-1}(y) = y_0 + \gamma \tan\left[\pi \left(F(y) - \frac{1}{2} \right)\right]$$

$$\Rightarrow F^{-1}(y) = \tan\left[\pi(y - \frac{1}{2})\right]$$

$y_0 = 0$
 $\gamma = 1$

Trobar c

$$C \approx \frac{f_y(y)}{g_y(y)}$$

$$h(y) = \frac{\frac{1}{\sqrt{2\pi}} e^{-y^2/2}}{\frac{1}{\pi(1+y^2)}} = \sqrt{\frac{\pi}{2}} (1+y^2) e^{-y^2/2}$$

Troblem es max.

$$\frac{\partial h(y)}{\partial y} = \cancel{\sqrt{\frac{\pi}{2}}} \sqrt{\frac{\pi}{2}} \left[2y e^{-y^2/2} + (1+y^2) e^{-y^2/2} (-y) \right]$$

$$= \sqrt{\frac{\pi}{2}} \left[\cancel{2y} 2y e^{-y^2/2} - y(1+y^2) e^{-y^2/2} \right]$$

$$= \sqrt{\frac{\pi}{2}} \left[y e^{-y^2/2} (2 - (1+y^2)) \right] =$$

$$= \sqrt{\frac{\pi}{2}} \left\{ y e^{-y^2/2} (1 - y^2) \right\}$$

$$= \sqrt{\frac{\pi}{2}} y e^{-y^2/2} (1 - y^2) = 0$$

$$\cancel{y(1-y^2)} = 0$$

$$\rightarrow y_1 = 0$$

$$\rightarrow 1 = y^2 \rightarrow y_2 = \sqrt{1} = \pm 1$$

Avaluem:

$$h(0) = \sqrt{\frac{\pi}{2}} \approx 1,253$$

$$h(-1) = \sqrt{\frac{\pi}{2}} \cdot 2 e^{-1/2} \approx 1,5203$$

$$h(1) = \sqrt{\frac{\pi}{2}} \cdot 2 \cdot e^{-1/2} \rightarrow$$

} es es max

$$\text{Així: } \left[C \approx 1,5203 \approx \sqrt{\frac{\pi}{2}} \frac{2}{\sqrt{e}} = \sqrt{\frac{2\pi}{e}} \right]$$